Bubonic Plague:

The plague caused by a bacterium (Yersinia Pestis) and characterized by especially by the formation of buboes is called bubonic plague. This Plague is currently managed monitoring for plague and reducing the number of rodenthosts for fleas.

Milk composition .5 marks

Milk used for bottling or drinking must by law contain 3.6% fat and 8.5% SNF. Creameries buy milk in many different ways – sometimes at a flat rate per kg. Sometimes, however, it may be by percentage butterfat or protein. Milk composition varies also amongst different breeds. The milk of the Jersey cow has the highest butterfat and SNF content of any cow. Also the composition of milk varies during milking. The milk at the start may only contain 1% fat, while the milk at the end ("The Stripping") may contain 10% fat.

Hygiene and Milk Quality:

A science of the establishment and maintenance of health is called hygiene and white liquid with balance diet is called milk. The hygiene and milk quality is as follows: The milk without dirt (bacteria or antibiotics, poor milk filters by bacteria or antibiotics, low cream and animal mastitis) is hygiene and milk quality.

Animal by-Products:

The animal by products are as follows:

1-Bones =Button, glue, mineral supplement for livestock feed (Ca)

2--Fat =Chemicals, salves, creams, dressings, lubricants, soaps, food

3- Glands =Medicines, food additives

4- Collagen=Glue, Gelatin

5- Intestinal & Stomach tissue =lunch meats, surgical sutures, strings for musical and sports instruments
6- Fertilizer=Natural

Bio accumulation:

Bioaccumulation: when a chemical accumulates in animal fat (historical fact) Bioaccumulation is the gradual accumulation of substances, such as pesticides, or other chemicals in an organism. Bioaccumulation occurs when an organism absorbs a substance at a rate faster than that at which the substance is lost by catabolism and excretion.

What is bio magnification.

Bio-magnification: when an organism accumulates residues at higher concentrations than the organisms they consume Some organisms accumulate chemical residues in higher concentrations than those found in their food source. This process is called biomagnification. For example, invertebrates with pesticides in their tissues are eaten by fish, which are then eaten by birds. The birds at the top of the food chain accumulate the highest concentration of pesticide residues. Pesticides that bioaccumulate and biomagnify have been removed from the marketplace.

Biomagnification, also known as bioamplification or biological magnification, is the increasing concentration of a substance, such as a toxic chemical, in the tissues of tolerant organisms at successively higher levels in a food chain.

Commonly cultured fish name:

• **Food fish** • Many species, Catfish, Tilapia, Rainbow trout, Atlantic Salmon, shrimp, Shellfish, Striped Bass, Others

Ornamental: fish –Aquaria –Backyard ponds

Natural stock enhancement: –Salmon –Trout –Black sea bass –Red Drum –Many others…

Bait fish: –Minnows –Shiners –Goldfish (carp)

Species Selection:

Producer's expertise • Water supply and climate • Species biology • Marketability • Production methods • Production economics.

Why practice IPM:

The IPM practice for to Maintains balanced ecosystems, pesticides alone may be ineffective, promotes a healthy environment, saves money and maintains a good public image.

Components of IPM:

1. Identify and Understand
2. Monitor the Pest
3. Develop the IPM Goal
4. Implement the IPM Program
5. Record and Evaluate Results

Define pullet:

"Young, immature female chicken less than 5-6 months of age"

Intrinsic AND extrinsic:

intrinsic incubation period (in vertebrate host). The intrinsic incubation period is **the time taken by an organism to complete its development in the definitive host**.

extrinsic incubation period (in arthropod vector). The extrinsic incubation period (EIP) of malaria, also called the period of sporogony, describes **the time it takes for parasites to develop in the mosquito from point of ingestion via an infected blood meal**.

Housefly mouth pars:

Fulcium, clypeus, rostrum, stipes, haustellum,theca, maxillarypalp, labrum, prementum, labellum, pseudotracea, furca and prestomal teeth are mouth parts of housefly.

Pest resistance:

The Insecticide Resistance Action Committee (IRAC) definespest resistance as "a heritable change in the sensitivity of apest population that is reflected in the repeated failure of a product to achieve the expected level of control when used according to the label recommendation for that pest species.

Natural pest control factors:

The natural pest control factors are as follows: 1-Wind 2-Temperature 3-Humidity, rain 4-Rivers, lakes, mountains 5- Pathogens, predators 6-Food supply of the pest.

Third-generation Pesticides:

1. Organophosphates and carbamates 2. Less persistent in environment (good deal) 3. Acutely potent nerve toxins 4. More lethal in low dose than organochlorines.

Monogastric and polygastric:

A monogastric organism has a simple single-chambered stomach, compared with a ruminant organism, like a cow, goat, or sheep, which has a four-chambered complex stomach. Monogastric (simple stomach) Pig, Dog, Human

Ruminants are mammals that are able to acquire nutrients from plant-based food by fermenting it in a specialized stomach prior to digestion, principally through microbial actions. The process, which takes place in the front part of the digestive system and therefore is called foregut fermentation, typically requires the fermented ingesta (known as cud) to be regurgitated and chewed again. The process of rechewing the cud to further break down plant matter and stimulate digestion is called rumination

Polygastric (ruminants) ′ Cow, Sheep, Goat

Treatment of mumbal foot. 5 marks:

a) Soak foot in warm water and Epsom salts. b) disinfect with alcohol. c) If skin is open, drain pus from abscess. d) Flush abscess cavity with hydrogen peroxide to cleanout pus and debris. e) Pack the cavity with antibiotic ointment. f) wrap the foot with gauze and elastic bandage.

Vector-Borne Diseases:

disease: Vectors 1-Dangue: Mosquito 2-Typhus: Rodent 3-Asthma: Cockroach 4-Bubonic Plague: Fleas 5- Skin infection: Louse.

Average milk composition:

Milk: a-Water 87.5% b-Total Milk Solids 12.5%

1-Fat 3.8% 2- Solids Non Fat (SNF) 8.7% i-Lactose 4.6% ii-Protein (Casien) 3.1% iii-N Protein 0.2% iv-Vitamins & Minerals 0.8

Hemimatabolous and holometabolous:

The gradual or incomplete metamorphosis is called hemimetabolous and complete metamorphous is called holometabolous.

OXYGENATION OF FISHES:

Oxygen first limiting factor in recirculation fish culture system ➢ O2 is consumed by fish and bacteria inhabiting bio filter ➢ In fish tank O2 should not be less than 5 ppm and in outgoing bio filter water not less than 2 ppm ➢ When these levels drop below these concentrations aeration becomes necessary.

Spring viremia crops:

Cause ✓ A viral infection caused by Rhabdovirus carpio. Symptoms ✓ Darkening of skin, pale gills, pop eye, protruding vent, bleeding in gills skin and eyes. ✓ Lethargy, abnormal swimming positions, sitting on bottom of the tank. **Treatment** ✓ No known treatment.

Mode of action of pesticide:

Some pesticides are systemic and actually move through animal or plant tissues and the pest encounters it when feeding on the animal or plant. Other pesticides work by contact action like battery acid on your pants. You must make sure that you apply the material so that it comes into direct contact with the pest.

Ari lac and Phunki lac:

Lac can be scraped from the twigs before or after the emergence of larvae. If it is used for manufacturing before the emergence of larvae, the type of lac produced is called as “Ari lac” and if it is used for manufacturing purpose after swarming of larvae has occurred, the lac is said to be “Phunki lac”.

Baaki Allah Khair Kry Ga Inshallah…..

BEST OF LUCK .